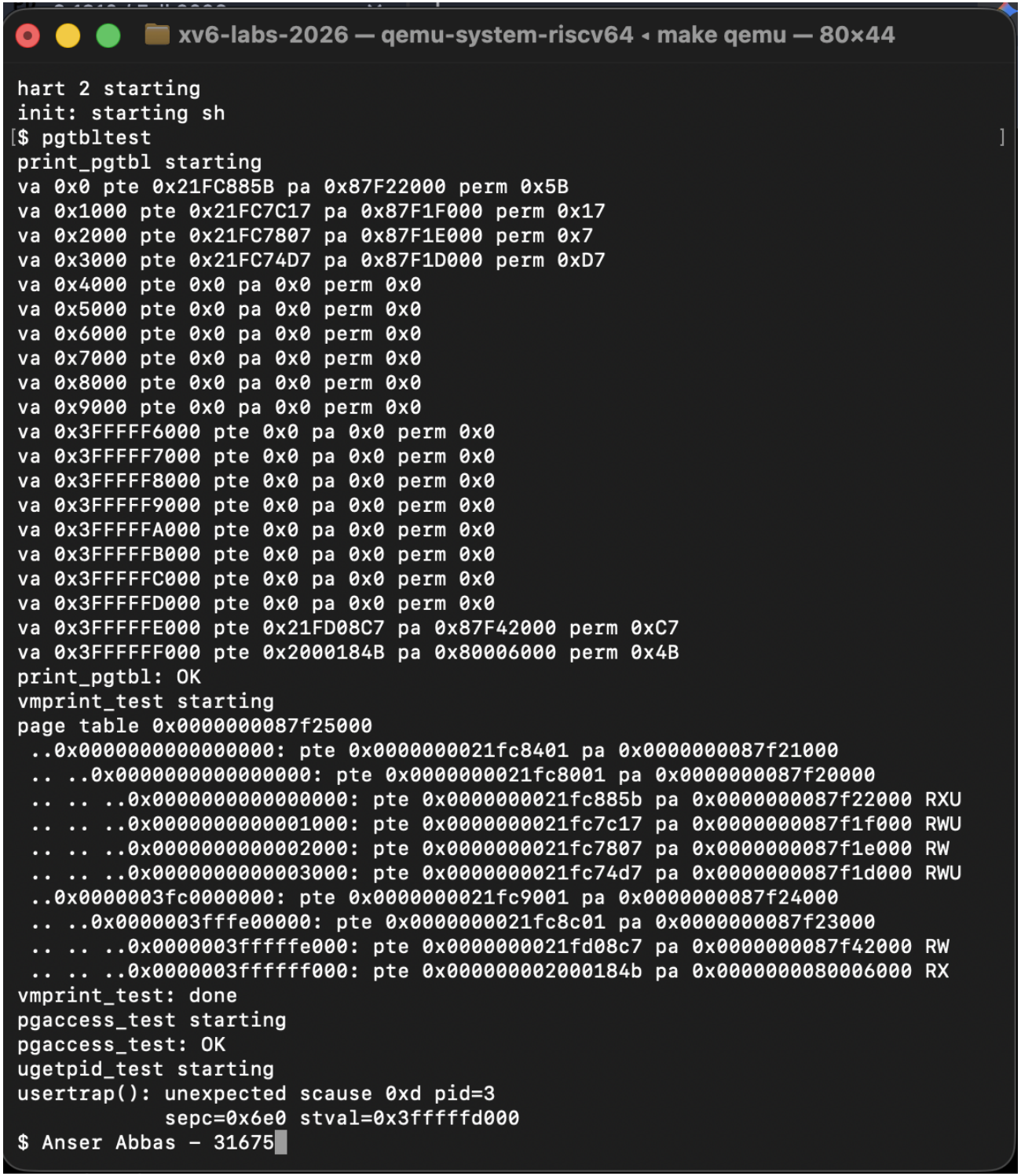
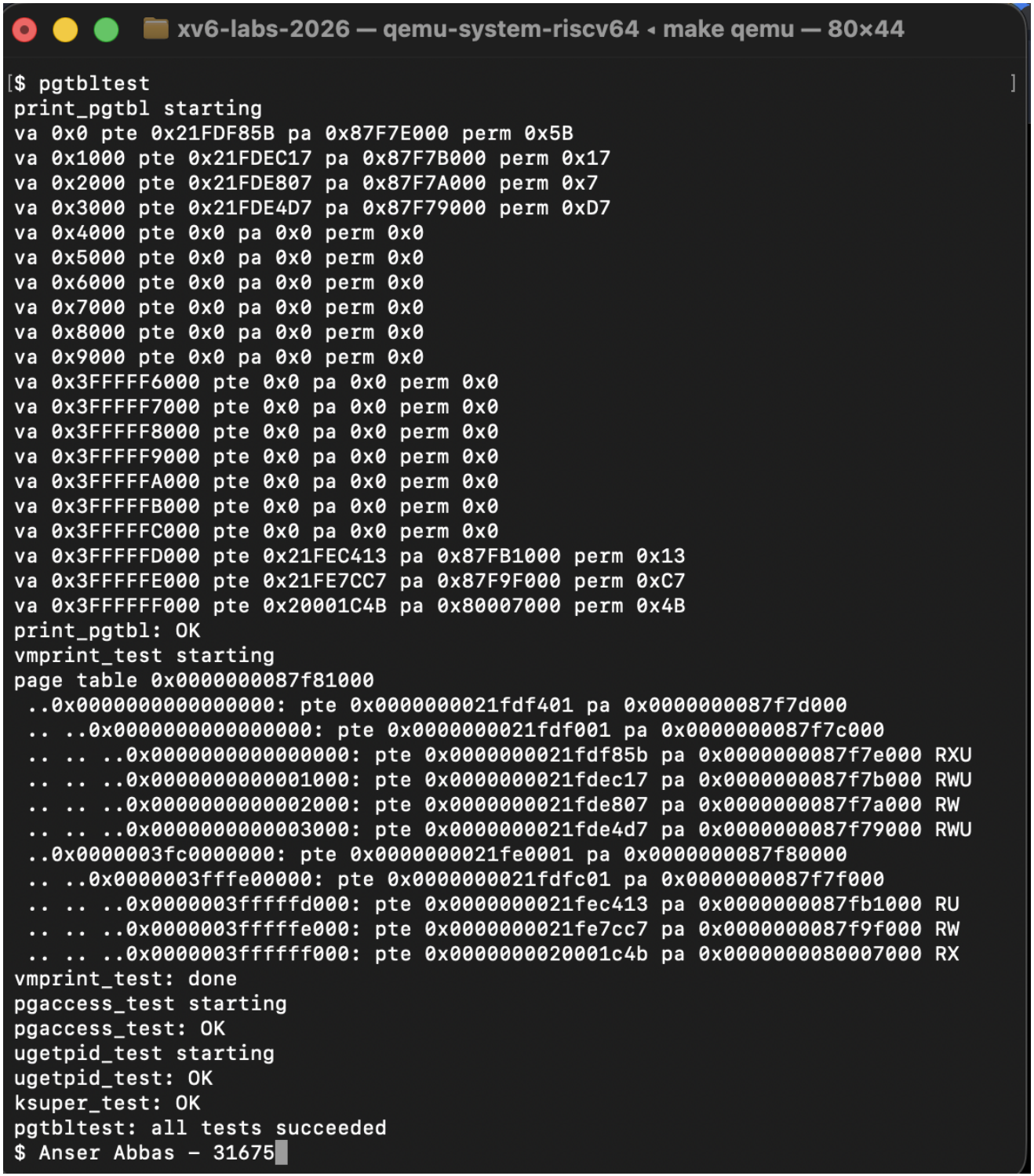

xv6-labs-2026 — qemu-system-riscv64 ◂ make qemu — 80×41
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
balloc: first 1047 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21FC885B pa 0x87F22000 perm 0x5B
va 0x1000 pte 0x21FC7C17 pa 0x87F1F000 perm 0x17
va 0x2000 pte 0x21FC7807 pa 0x87F1E000 perm 0x7
va 0x3000 pte 0x21FC74D7 pa 0x87F1D000 perm 0xD7
va 0x4000 pte 0x0 pa 0x0 perm 0x0
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFD000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFE000 pte 0x21FD08C7 pa 0x87F42000 perm 0xC7
va 0x3FFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
vmprint_test starting
vmprint_test: done
pgaccess_test starting
pgtbltest: pgaccess_test failed: incorrect access bits set, pid=3
$


xv6-labs-2026 — qemu-system-riscv64 ◂ make qemu — 80×44
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21FC885B pa 0x87F22000 perm 0x5B
va 0x1000 pte 0x21FC7C17 pa 0x87F1F000 perm 0x17
va 0x2000 pte 0x21FC7807 pa 0x87F1E000 perm 0x7
va 0x3000 pte 0x21FC74D7 pa 0x87F1D000 perm 0xD7
va 0x4000 pte 0x0 pa 0x0 perm 0x0
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFD000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFE000 pte 0x21FD08C7 pa 0x87F42000 perm 0xC7
va 0x3FFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
vmprint_test starting
page table 0x0000000087f25000
..0x0000000000000000: pte 0x0000000021fc8401 pa 0x0000000087f21000
.. ..0x0000000000000000: pte 0x0000000021fc8001 pa 0x0000000087f20000
.. .. ..0x0000000000000000: pte 0x0000000021fc885b pa 0x0000000087f22000 RXU
.. .. ..0x0000000000001000: pte 0x0000000021fc7c17 pa 0x0000000087f1f000 RWU
.. .. ..0x0000000000002000: pte 0x0000000021fc7807 pa 0x0000000087f1e000 RW
.. .. ..0x0000000000003000: pte 0x0000000021fc74d7 pa 0x0000000087f1d000 RWU
..0x0000003fc0000000: pte 0x0000000021fc9001 pa 0x0000000087f24000
.. ..0x0000003fffe00000: pte 0x0000000021fc8c01 pa 0x0000000087f23000
.. .. ..0x0000003fffffe000: pte 0x0000000021fd08c7 pa 0x0000000087f42000 RW
.. .. ..0x0000003ffffff000: pte 0x000000002000184b pa 0x0000000080006000 RX
vmprint_test: done
pgaccess_test starting
pgtbltest: pgaccess_test failed: incorrect access bits set, pid=3
$


xv6-labs-2026 — qemu-system-riscv64 ◂ make qemu — 80×44
hart 2 starting
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21FC885B pa 0x87F22000 perm 0x5B
va 0x1000 pte 0x21FC7C17 pa 0x87F1F000 perm 0x17
va 0x2000 pte 0x21FC7807 pa 0x87F1E000 perm 0x7
va 0x3000 pte 0x21FC74D7 pa 0x87F1D000 perm 0xD7
va 0x4000 pte 0x0 pa 0x0 perm 0x0
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFD000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFE000 pte 0x21FD08C7 pa 0x87F42000 perm 0xC7
va 0x3FFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
vmprint_test starting
page table 0x0000000087f25000
..0x0000000000000000: pte 0x0000000021fc8401 pa 0x0000000087f21000
.. ..0x0000000000000000: pte 0x0000000021fc8001 pa 0x0000000087f20000
.. .. ..0x0000000000000000: pte 0x0000000021fc885b pa 0x0000000087f22000 RXU
.. .. ..0x0000000000001000: pte 0x0000000021fc7c17 pa 0x0000000087f1f000 RWU
.. .. ..0x0000000000002000: pte 0x0000000021fc7807 pa 0x0000000087f1e000 RW
.. .. ..0x0000000000003000: pte 0x0000000021fc74d7 pa 0x0000000087f1d000 RWU
..0x0000003fc0000000: pte 0x0000000021fc9001 pa 0x0000000087f24000
.. ..0x0000003fffe00000: pte 0x0000000021fc8c01 pa 0x0000000087f23000
.. .. ..0x0000003fffffe000: pte 0x0000000021fd08c7 pa 0x0000000087f42000 RW
.. .. ..0x0000003ffffff000: pte 0x000000002000184b pa 0x0000000080006000 RX
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
usertrap(): unexpected scause 0xd pid=3
sepc=0x6e0 stval=0x3fffffd000
$ Anser Abbas - 31675


xv6-labs-2026 — qemu-system-riscv64 ◂ make qemu — 80×44
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21FDF85B pa 0x87F7E000 perm 0x5B
va 0x1000 pte 0x21FDEC17 pa 0x87F7B000 perm 0x17
va 0x2000 pte 0x21FDE807 pa 0x87F7A000 perm 0x7
va 0x3000 pte 0x21FDE4D7 pa 0x87F79000 perm 0xD7
va 0x4000 pte 0x0 pa 0x0 perm 0x0
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFD000 pte 0x21FEC413 pa 0x87FB1000 perm 0x13
va 0x3FFFFFE000 pte 0x21FE7CC7 pa 0x87F9F000 perm 0xC7
va 0x3FFFFFF000 pte 0x20001C4B pa 0x80007000 perm 0x4B
print_pgtbl: OK
vmprint_test starting
page table 0x0000000087f81000
..0x0000000000000000: pte 0x0000000021fdf401 pa 0x0000000087f7d000
.. ..0x0000000000000000: pte 0x0000000021fdf001 pa 0x0000000087f7c000
.. .. ..0x0000000000000000: pte 0x0000000021fdf85b pa 0x0000000087f7e000 RXU
.. .. ..0x0000000000001000: pte 0x0000000021fdec17 pa 0x0000000087f7b000 RWU
.. .. ..0x0000000000002000: pte 0x0000000021fde807 pa 0x0000000087f7a000 RW
.. .. ..0x0000000000003000: pte 0x0000000021fde4d7 pa 0x0000000087f79000 RWU
..0x0000003fc0000000: pte 0x0000000021fe0001 pa 0x0000000087f80000
.. ..0x0000003fffe00000: pte 0x0000000021fdfc01 pa 0x0000000087f7f000
.. .. ..0x0000003fffffd000: pte 0x0000000021fec413 pa 0x0000000087fb1000 RU
.. .. ..0x0000003fffffe000: pte 0x0000000021fe7cc7 pa 0x0000000087f9f000 RW
.. .. ..0x0000003fffffff000: pte 0x0000000020001c4b pa 0x0000000080007000 RX
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
ugetpid_test: OK
ksuper_test: OK
pgtbltest: all tests succeeded
$ Anser Abbas - 31675